integration tests, static code analysis, and packaging. At the end of this stage, an executable application artifact is published to an artifact repository. That is a *release candidate*. Chapter 3 covered all the main steps except for the final packaging and publishing of a release candidate. That's what you'll see in this section.

6.4.1 Building release candidates in the commit stage

After running static code analysis, compilation, unit tests, and integration tests, it's time to package the application as an executable artifact and publish it. In our case, the executable artifact is a container image that we'll publish to a container registry.

An essential idea in continuous delivery, also present in the 15-Factor methodology, is that you should build artifacts only once. At the end of the commit stage, we'll produce a container image that we can reuse in any following stage in the deployment pipeline up to production. If the pipeline proves something is wrong (a test fails) at any point, the release candidate is rejected. If the release candidate goes through all subsequent stages successfully, it's proven to be ready for deployment in production.

After we build an executable artifact, we can perform additional operations before publishing it. For example, we could scan it for vulnerabilities. That's what we're going to do with grype, much as we did for the codebase. A container image includes application libraries but also system libraries that were not included in the previous security analysis. That's why we need to scan both the codebase and the artifact for vulnerabilities. Figure 6.15 illustrates the new steps we will add to the commit stage for building and publishing a release candidate.